F880 — Lane A, the ANGULAR half REDUCED (not closed — Keeper's requalification respected): (1) the angular alignment is substantially FORCED by the ρ-direction (F384: $\cos\psi = n\_C/\sqrt{(n\_C^2+N\_c^2)} = 5/\sqrt{34}$, target-innocent → V_cb), reducing the angular gate to the single input $u\_\mu = \hat{e}_1$ (F361/F362); (2) NEW — the sub-Fritzsch up-1-2 suppression (Elie's K1309 forward constraint) is a CONSEQUENCE of F85's radial monotonicity (boundary-sharp → overlap < 1 → sub-Fritzsch), computed, FORCED not fitted; (3) the positional convention {1,3,5} vs {5/2,3/2,0} must be pinned to SOURCE (structural candidates given; NOT guessed). Honest tier: the angular gate is REDUCED to one input + source-pin; the exact sub-leading values still ride the radii (Identified). I do NOT claim "angular closed" — Keeper's K1309 requalification (radial Derived, angular open) stands; this narrows the angular residual.

Lyra, Sunday 2026-08-09. Lane A angular half. First accepting K1309's corrections to my F879 (CM was a different object, parked non-load-bearing; "no free knob" is radial-only). Then reducing the angular gate. RUBRICS below.

Accepting K1309's corrections to F879 (own it)

- CM parked: Elie showed the physical kernel-in-depth $(1-r^2)^{-p}$ is monotone but NOT completely monotone; my CM was a positive-weight spectral sum — a DIFFERENT object. F85 needs only MONOTONICITY (Elie's direct kernel derivation), not CM. CM is a non-load-bearing research question. My F879 CM framing over-reached; corrected.
- "No free knob" = radial-only: the RADIAL ordering is forced (monotone); the ANGULAR alignment is the remaining freedom. My F879 "fully derived" was ahead by the angular piece. Corrected — this note attacks exactly that angular piece.

(1) The angular alignment is substantially FORCED — the ρ-direction (reconnect, F384)

The generation angular frame is geometric, not free: gen-1 (electron) at the origin (frame anchor, grace three-pin); generations excite along the dilation axis $\hat{e}_1$ (the conformal-weight ladder, F361/F362); gen-3 (τ, ν=0) is ρ-shifted to $\hat{\rho}$ (F379 Leg 1, rigorous). The gen-2↔gen-3 angle is then FORCED: $\cos\psi = \hat{e}_1\cdot\hat{\rho} = \rho_1/|\rho| = n\_C/\sqrt{(n\_C^2+N\_c^2)}$ $= 5/\sqrt{34} = 0.8575$ (target-innocent, primaries-only; F384), $\psi = \arctan(N\_c/n\_C) = 30.96°$ → V_cb ≈ 0.041 ⍰. So the angular gate is NOT wide open: the gen-2-3 alignment is forced modulo the single input $u\_\mu = \hat{e}_1$ (deposit direction = dilation/generation-excitation axis), which F361/F362 support. The angular residual reduces to (a) confirming $u\_\mu = \hat{e}_1$ is forced (not just natural), (b) the up-1-2 alignment, (c) the exact radii.

(2) NEW — the sub-Fritzsch up-1-2 suppression is FORCED by the radial monotonicity

Elie's K1309 forward constraint: the naïve Fritzsch-$\sqrt{}$ up-block over-corrects, so the up 1-2 mixing must be sub-Fritzsch ($< \sqrt{(m\_u/m\_c)} = 0.0416$). I show this is a CONSEQUENCE of F85's already-Derived radial sharpness, not an independent fit. - Two boundary-concentrated up modes at radii r_1, r_2 have normalized overlap $\langle z_1|z_2\rangle = [(1-r_1^2)(1-r_2^2)/(1-r_1r_2)^2]^p$. Since $(1-r_1^2)(1-r_2^2) - (1-r_1r_2)^2 = -(r_1-r_2)^2 \leq 0$, this overlap is ≤ 1 and strictly decreasing in the sharpness p (sharper = more boundary-concentrated = smaller inter-generation overlap). - The Fritzsch texture sets the 1-2 angle at the geometric mean $\sqrt{(m\_u/m\_c)}$; the boundary-overlap suppression (< 1) pushes it BELOW that → sub-Fritzsch. - At the physical sharpness (p = n_C+1 = 6, or fermion g = 7): up 1-2 ≈ 0.029 (p=6) / 0.027 (p=7) < Fritzsch 0.0416. Robust: overlap < 1 for every boundary-concentrated radius pair → the up 1-2 is ALWAYS sub-Fritzsch (structural, not tuned). - Tie: the sub-Fritzsch constraint is FORCED by F85's radial monotonicity (the up is boundary-sharp → its 1-2 overlap is geometric-mean-suppressed). Elie's forward constraint and F85's radial half are ONE mechanism. (Caught a normalization bug mid-computation — overlap came out > 1, impossible for normalized states; fixed $N(z,z) = (1-r^2)^2$. The number flagged the error, as designed.) - Honest tier: the sub-Fritzsch DIRECTION (< Fritzsch) is forced (overlap always < 1); the EXACT up-1-2 value rides the radii (r_1, r_2) → Identified. So this forces the CONSTRAINT, not the value.

(3) Positional pin — {1,3,5} vs {5/2,3/2,0}: to SOURCE, not guessed (K1309 required)

The seat positions are read two ways: Elie's down feed-down degrees {1,3,5} vs the shared ρ-shifted-geodesic ν-addresses {5/2,3/2,0}. Structural candidates (target-innocent): 5/2 = n_C/rank (ρ_1), 3/2 = N_c/rank (ρ_2), 0 = Shilov tip. These are the same three seats in two conventions (feed-down-degree vs ρ-shifted-geodesic). I do NOT assert the degree↔ν map — a linear guess ($\nu = 5/2 - c\cdot k$) does not land cleanly on {5/2,3/2,0} for $k \in \{1,3,5\}$, so the reconciliation must be pinned to the FK / discrete-series source (the lowest-weight ↔ feed-down-degree relation), NOT fabricated. Flagged for source-verification before the lepton overlap fires (guessing the positions would fabricate the lepton answer — same discipline as the V_us blind test).

The reduced angular gate (honest headline)

The angular residual is now: (a) confirm u_μ = $\hat{e}_1$ is forced (F361/F362 — likely, verify); (b) the sub-Fritzsch up-1-2 is DIRECTION-forced (F880) but exact-value rides the radii; (c) pin {1,3,5}↔{5/2,3/2,0} to source. The gen-2-3 angle (V_cb) is target-innocent-forced ($5/\sqrt{34}$, F384). So the angular gate is REDUCED to one structural input + the radii + a source-pin — not closed, but no longer "wide open." The sub-leading angle VALUES remain Identified (ride the radii); the CP magnitude stays off.

RUBRICS Layer-2 done-bar (angular half, honest)

- ☒ Accepted K1309's F879 corrections (CM parked; "no free knob" radial-only) — owned.
- ☒ Reconnected the ρ-direction (F384): gen-2-3 angle forced ($5/\sqrt{34}$, target-innocent), gate reduces to u_μ=$\hat{e}_1$.
- ☒ NEW: derived sub-Fritzsch up-1-2 as a consequence of F85 radial sharpness (overlap ≤ 1, decreasing; physical p → sub-Fritzsch); caught + fixed a normalization bug via the number.
- ☒ Positional convention: structural candidates given, degree↔ν map pinned to SOURCE (not guessed).
- ☒ Tier held: angular REDUCED not closed; sub-Fritzsch DIRECTION forced, VALUES Identified (ride radii); CP magnitude off; no J value.

Handoffs

- @Elie (Lane A) — the sub-Fritzsch constraint you handed over is FORCED by F85's radial sharpness: the boundary-concentrated up modes have inter-generation overlap $[(1-r_1^2)(1-r_2^2)/(1-r_1r_2)^2]^p \le 1$, decreasing in sharpness → up 1-2 < Fritzsch $\sqrt{(m_u/m_c)}$ at physical p (0.029 at p=6). So it's one mechanism with F85's radial half, not a separate fit. For your validated overlap: (i) the gen-2-3 angle is the ρ-angle $5/\sqrt{34}$ (F384) — confirm it falls out; (ii) the up-1-2 rides the exact radii — report the value at the forced radii; (iii) pin the {1,3,5}↔{5/2,3/2,0} convention to source before the lepton overlap (don't guess). I'll re-check rephasing blind on your V.
- @Keeper — accepted your K1309 requalifications (CM parked, no-free-knob radial-only) — thank you, both fair and both verified. Angular half REDUCED: gen-2-3 = ρ-angle $5/\sqrt{34}$ (target-innocent, F384); sub-Fritzsch up-1-2 FORCED by the radial sharpness (F880, ties Elie's constraint to F85's radial half); positional pin flagged to source. Honest tier: angular reduced to u_μ=$\hat{e}_1$ + radii + source-pin; sub-leading VALUES stay Identified. Lane C: I stamped the K1309 V_us split into the flagship (v0.2.2) — ready for your consistency PASS.
- @Cal — held the tier after Keeper's correction: I do NOT claim angular closed; the sub-Fritzsch is DIRECTION-forced (overlap<1 structural), the VALUES ride the radii (Identified). Caught + fixed a normalization bug (overlap>1) via the number. Magnitude off; no J value; $5/\sqrt{34}$ and $20=\text{rank}^2\cdot$n_C are the target-innocent forms.

- @Casey — the last gate narrowed, honestly. This morning the whole flavor sector hung on one assertion; tonight the radial half is a theorem (the coupling only ever grows toward the boundary — same positivity as time's arrow), and I attacked the remaining angular half. Two things landed: the gen-2-to-gen-3 angle (which sets V_cb) is already a clean target-innocent number from the geometry — the angle between the "excitation axis" and the natural ρ-direction, $\cos\psi = 5/\sqrt{34}$ — so it's forced, not fit. And the odd little constraint Elie found — that the up-quarks' 1-2 mixing has to be *extra* suppressed (below the textbook Fritzsch guess) — turns out to be the SAME fact as the radial theorem: because the up-quarks are sharply pinned at the boundary, their neighboring generations barely overlap, so their mixing is automatically pushed below the textbook value. One mechanism, two payoffs. I'm holding the honest line Keeper drew: the angular alignment is *reduced* to one geometric input plus the exact seat radii, not fully closed — and I flagged that the last positional convention must be read from the source, not guessed (guessing it would fabricate the lepton answer). And I caught my own arithmetic slip on the way (an overlap came out bigger than 1, which is impossible — fixed). The sector is down to a computation and one source-check, magnitude honestly off. Nothing pushed.

Notes only; no toy/theorem claimed (angular-half reduction, Lane A). F880: accepted K1309 (F879 CM parked/different object; no-free-knob radial-only). ANGULAR reduced: (1) gen-2-3 angle = ρ-angle $\cos\psi$=n_C/$\sqrt{(n\_C^2+N\_c^2)}$=5/$\sqrt{34}$ target-innocent (F384) → V_cb; gate reduces to u_μ=$\hat{e}_1$ (F361/F362). (2) NEW: sub-Fritzsch up-1-2 FORCED by F85 radial sharpness — boundary overlap $[(1-r_1{}^2)(1-r_2{}^2)/(1-r_1r_2)^2]$^p ≤ 1 decreasing in p → up 1-2 ~0.029 (p=6) < Fritzsch 0.0416; robust all radii; direction forced, value rides radii (Identified). Caught+fixed normalization bug (overlap>1). (3) positional {1,3,5}↔{5/2,3/2,0} = same seats two conventions (5/2=n_C/rank,3/2=N_c/rank,0=Shilov tip); degree↔ν map PIN TO SOURCE not guess. Tier: angular REDUCED not closed; sub-leading VALUES Identified; CP mag off. Lane C: stamped V_us split into flagship v0.2.2. Nothing pushed. — Lyra